

HAND-OFF · FIRST PRINCIPLES THEN RECIPES · 18 AUGUST 2026 · EUROPE/BERLIN (PT)

JFTSE SeaWave — moving the wave in four directions (2026-08-18)

Author ThewindMom / JFTSE. Facts only. Unconfirmed items are marked **UNCONFIRMED** / **RECIPE** / **UNKNOWN**. No invented hashes, addresses, or live results. Do not commit or push from this note. Live exe + hashes re-read on this box ~19:40 PT.

One sentence

UseSkill has **no heading**. Travel is a client first-frame direction vector, then normalize, then $\text{pos} += \text{dir} * \text{speed} * \text{dt}$. Which component is large after normalize is the travel axis. The sign is which way. The other horizontal component is the 11-crest spread.

Status at a glance

Dir	Screen	Travel axis	Status
LTR left → right	left → right	+client X / script +X	CONFIRMED 17:43 PT 18 Aug 2026 by Thewind. Live cave + tape.
RTL right → left	right → left	−client X / script −X	RECIPE , not live-confirmed. Same cave, float −1.0. Not taped tonight.
UTD far → player	top → bottom (toward camera)	−client Z / script −Y	VANILLA AXIS . Already travels this axis. Which way = attacker Z. Forced-cave sign UNCONFIRMED .
DTU player → far	bottom → top (away from camera)	+client Z / script +Y	VANILLA AXIS , opposite dir.z. Forced-cave sign UNCONFIRMED .

UNKNOWN — do not guess live

RTL live look. Forced UTD/DTU cave signs on tape. Whether flipping attacker is enough for vanilla UTD vs DTU without a cave. Dummy-4 world position was not dumped.

1. First principles

The UseSkill packet has **no heading**. Live Java constructor:

```
S2CMatchplayUseSkill(byte attacker, byte target, byte skillId, byte seed,
    float xTarget, float zTarget, float yTarget)
```

It writes those seven fields and stops. No yaw, no travel enum, no direction byte. What travels is entirely a **client first-frame direction vector**, then normalize, then integrate. From the 18 August RE note (evidence/seawave-ltr/README.md):

- Consume 0x4EF1E0 copies packet skillId, skill object, ShotSpeed, ShotRot. Targeting 12/13 copies the three floats into instance+0x4C/+0x50/+0x54 and sets +0x58=1. Still no heading float.
- Tick 0x4F0CA0 walks the projectile list and calls mover 0x4E9CE0 (ShotType 3/6/7 only).
- First frame** (skillId 27 at cmp [ebp+8], 0x1B / VA 0x4E9D91) builds dir for shot index 0..10.
- D3DXVec3Normalize writes the unit vector into projectile+0x54/+0x58/+0x5C.
- Later ticks skip the index switch (projectile+0x10 != 0 → 0x4EA133) and **freeze** that dir.
- Every tick at 0x4EA2CD: $\text{pos} += \text{dir} * (\text{projectile} + 0x50) * \text{dt}$. projectile+0x50 is ShotSpeed + index*35 on first frame for skill 27 (RE note; not re-debugged tonight).

Planted xyz is the **drop**. With ShotRot=0 it is not a steering target. Changing xyz never flips heading.

Two levers (this is the whole trick)

#	Lever	What it does
1	Which component is large after normalize	Travel axis
2	Sign of that component	Which way on that axis
3	The other horizontal component	11-crest spread (wall width or fan)

Vanilla: |dir.z| dominates → travel baseline↔net; dir.x = −0.5...+0.5 is a full-width wall along X. LTR cave: |dir.x| dominates (+1.0) → travel sideline↔sideline; old X-spread moves onto dir.z → crests fan along the court. You cannot get a sideways wall from XML, ShotRot, TPosition, or a cloned skill id. The first-frame build is hardcoded to **packet skillId 27**.

Worked normalize (arithmetic, not a live dump)

Attacker Z was not dumped, so vanilla |dir.z| is unknown. The point is the ratio. If |dir.z| is large versus the 0.5 spread, normalize leaves travel on Z and shrinks the X-spread into a unit vector. The LTR numbers below use only the documented

floats (+1.0 travel, -0.5...+0.5 fan). They are math, not a taped measurement.

Setup	Before normalize (one crest)	After normalize	What you see
Vanilla, Z ■ 0.5	$\text{dir} \approx (\text{spread}, 0, \text{attackerZ})$	$\text{dir.z} \approx \text{sign}(\text{attackerZ})$. $\text{dir.x} \approx \text{spread}/ Z $ (small)	Full-width wall. Travel baseline↔net. Sign = attacker.
LTR center crest	(+1.0, 0, 0)	(+1.0, 0, 0)	Pure +X. Band travels left → right.
LTR edge crest	(+1.0, 0, ±0.5) length $\sqrt{1.25} \approx 1.118$	(+0.894, 0, ±0.447)	Still mostly +X. Z-fan is why foam can show at both baselines. Do not zero Z.
RTL edge crest (recipe)	(-1.0, 0, ±0.5)	(-0.894, 0, ±0.447)	Same fan, travel -X. Not taped tonight.
Forced UTD (recipe)	(spread, 0, -1.0)	mostly -Z, small X	Full-width wall toward player. Sign inferred, not taped.
Forced DTU (recipe)	(spread, 0, +1.0)	mostly +Z, small X	Full-width wall toward far baseline. Sign inferred, not taped.
Rejected Z-zero LTR	(+1.0, 0, 0) for every crest	(+1.0, 0, 0) for every crest	Thin net-line. Thewind rejected this look.

Registers at the cave site (VA 0x4EA111): $\text{edx} = \text{dir.x}$, $\text{eax} = \text{dir.y}$, $\text{ecx} = \text{dir.z}$. Stores: $[\text{esp}+0x18]=\text{edx}$, $[\text{esp}+0x1C]=\text{eax}$, $[\text{esp}+0x20]=\text{ecx}$, then `jmp 0x4EA125`. LTR/RTL rewrite edx (travel on X) and copy old edx into ecx (fan on Z). A forced UTD/DTU cave would leave edx alone (keep the X-spread) and rewrite ecx only.

2. Coordinate systems (do not mix them)

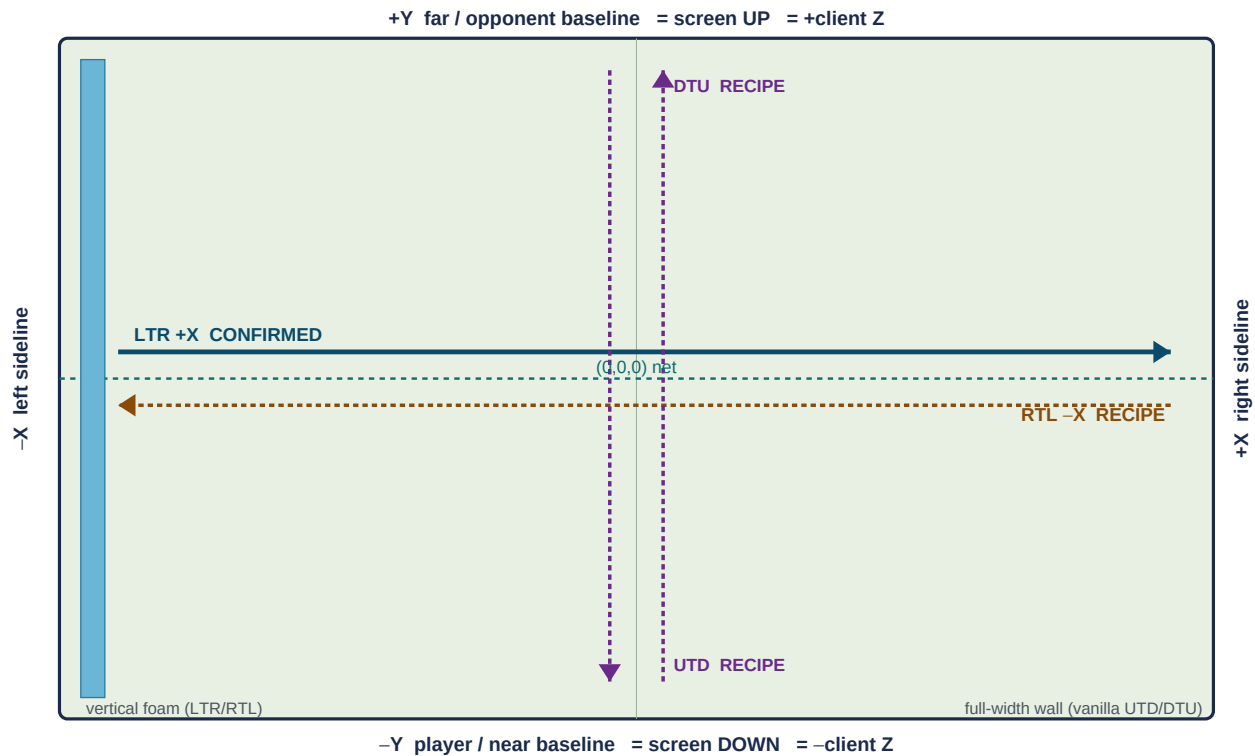
Client world is D3D Y-up. Court constants from notes/guardian-shot-ai-re.md: net Z=0, baselines about ± 115 , sidelines about ± 55 . Origin-findings live drops at X= ± 150 already sat on the visible sideline; Y= ± 150 sat on the baselines. Sideline saturation: x=-200 is already past the left edge, so x=-500 looks the same (confirmed tonight for LTR).

Name	Axis	Meaning	Scale (notes)
client X	sidelines	screen left / right	$\sim \pm 55$
client Y	height	up from the court; not a SeaWave travel axis	—
client Z	baselines	along the court; net = 0	$\sim \pm 115$

Packet floats xTarget, zTarget, yTarget and the JFTSE script names are a rename trap:

Packet	= client	= script	Screen (player at bottom, looking at opponent)
xTarget	client X	step.x	Left = -X. Right = +X.
zTarget	client Y (height)	step.z	Height. 0 for every confirmed / recipe spawn here.
yTarget	client Z (depth)	step.y	Up the screen = toward opponent = +Y (script) = +client Z. Down = toward player = -Y = -client Z.

wavetest.js builds new S2CMatchplayUseSkill(attacker, 4, 27, seed, step.x, step.z, step.y). Chat labels print xyz=(x, z, y) in that order. Origin-findings (vanilla, live): (0,0,0) is **net-center**, not under the player. +Y = far / opponent baseline. -Y = player / near baseline. Dummy 4 and dummy 5 both fire. Player at -Y looking +Y: **+X is their right**. LTR with dir.x = +1.0 was taped traveling left → right, so that screen mapping is live-confirmed.



Schematic, not to scale. Solid blue arrow = LTR confirmed. Dashed amber = RTL recipe. Dashed purple = vanilla-axis recipes (UTD down the page, DTU up the page). LTR/RTL foam is a vertical band; vanilla UTD/DTU foam is a full-width wall.

3. Vanilla first-frame (skillId 27 only)

```
dir.x = hardcoded spread -0.5, -0.4, ..., 0, ..., +0.5 (11 crests)
dir.y = FPU leftover (~0)
dir.z = [esp+0x30] ; attacker-related Z from 0x524C30
then D3DVec3Normalize into projectile+0x54 / +0x58 / +0x5C
```

Normalize of (small X, 0, large Z) is why vanilla foam is a **full-width band traveling baseline↔net**. ShotCnt=11 is the 11 crests. FieldItem XML (FieldItem_Skills_Ini3.xml ID 27 Name SeaWave), read tonight:

Field	Value	Notes
ID / Name	27 / SeaWave	Packet skillId 27. Cave keys on 0x1B.
ShotType	6	Mover 0x4E9CE0 handles types 3/6/7.

Field	Value	Notes
ShotCnt	11	Hardcoded 11-crest spread for skillId 27.
Targeting	13	UseSkill xyz → instance+0x4C/+0x50/+0x54. No heading.
TPosition	0, 0, 150	Unused for ShotType 6. Matches baked knockback, not motion.
ShotSpeed	100.0	projectile+0x50 = this + index*35 (RE note).
ShotRot	0.0	Turn-rate deg/s, not yaw. 0 ⇒ dir never steers.

Origin-findings also wrote "Skill table id 28". That wording is **not re-verified** against a non-FieldItem Skill table tonight. Packet skillId is 27. Use 27.

Why a FieldItem clone cannot go sideways

- The 11-piece spread and this dir build run only when **packet skillId == 27**. A new skill id **loses** the 11-crest spread.
- TPosition is unused for ShotType 6 (falls through to a zero vec at spawn 0x4EF726).
- Changing ShotRot / TPosition in server XML does not rotate SeaWave. Client table is encrypted Res/Script/PubETC/Ini3.res → Fielditem_INI3.set.
- ShotRot=90 would only enable steering toward dest; it would not rotate the wall.

ShotRot is turn-rate, not yaw

At 0x4EA1A7: $dt * \text{ShotRot (degrees/second)}$ vs $\text{acos}(\text{dot}(\text{dir}, \text{dest-pos})) * 57.29577$, then rotate by that many degrees converted to rad (0.017453, fsincos). Dest is the target-player pos (Y forced 0 for skill 27) unless targeting 12/13, in which case dest is the planted xyz. SeaWave ShotRot=0 → max turn 0 → **dir never steers**. Planted xyz does not aim the wave.

Knockback X is still forced 0 for skill 27 at 0x4F1C91 (not patched). Expect travel-axis motion with the old along-court pop. LTR knockback feel was **not** re-measured tonight.

4. Left → right (LTR) — CONFIRMED

Thewind confirmed this look 17:43 PT / 15:43 UTC, 18 August 2026
Foam is a **vertical band** traveling sideline → sideline (left → right). Crests fan along the court — foam can show at both baselines. That fan is the look. **Do not xor ecx (31 C9)**. Thewind said the Z-zero look is wrong.

Cave (skillId 27 only), after edx/eax/ecx = (dir.x, dir.y, dir.z) at VA 0x4EA111:

<pre>cmp [ebp+8], 0x1B ; skillId 27 jne +7 ; other skills: original edx/eax/ecx stores mov ecx, edx ; 8B CA dir.z = old dir.x (-0.5 ... +0.5 fan) mov edx, 0x3F800000 ; BA 00 00 80 3F dir.x = +1.0 (LTR) mov [esp+0x18], edx ; dir.x mov [esp+0x1C], eax ; dir.y (unchanged) mov [esp+0x20], ecx ; dir.z jmp 0x4EA125 ; normalize + integrate</pre>	
Item	Value (verified on live exe ~19:40 PT)
Site VA / file	VA 0x4EA115 / file 0x0EA115
Site bytes	E9 3E BB 1F 00 90×9 → jmp 0x6E5C58
Cave VA / file	VA 0x6E5C58 / file 0x2E5C58 (30 bytes, .text raw padding)
Cave bytes	83 7D 08 1B 75 07 8B CA BA 00 00 80 3F 89 54 24 18 89 44 24 1C 89 4C 24 20 E9 AF 44 E0 FF
.text VirtualSize	file 0x220: was 0x2E4C58 → now 0x2E4C80. SectionAlignment 0x1000 already maps the page. SizeOfImage unchanged.
Index-fail path	0x4EA123 dd d8 (fstp st0) is not touched
Live sha256	cf551df8bd42f32676f8b01e54496a1b74473c90ddfa1354288f6545dea92f7c

Confirmed spawn: **5 waves, 5 s apart, attacker=4, xyz=(-200, 0, 0)** “net center, further left”. Packet skillId 27. Target byte 4. Seed random 0..126.

```
{ attacker: 4, x: -200, z: 0, y: 0, guess: "net center, further left" }    × 5
delay = 500 + i * 5000 ms
new S2CMatchplayUseSkill(4, 4, 27, seed, -200, 0, 0)
```

Sideline scale ~±55: x=-200 is already off the left edge. x=-500 looks the same (saturation), not a swapped axis. x=-500 was **not** the confirmed take (ltr-neg500.mp4 exists, 17:45 PT; Thewind has not confirmed -500).

Rejected LTR experiments — do not “improve” back into these

Attempt	What it was	Why it is wrong
Z-spread-zero cave	xor ecx, ecx (31 C9). SHA 967cc05871330521c9c7dc0ade4f9f5989fce5797d81362174c10c985ff71f3f. On disk as FantaTennis.exe.ltr and in apply_ltr_patch.py / bytes.txt.	Thewind 17:15–17:28 PT: foam no longer looked like the first tape. Restored 31 C9 → 8B CA at 17:29 PT (first-ltr-restore.txt).
Spawn x = +200	ltr-player-right*.mp4 (16:52–17:09 PT).	Wrong side. First take starts on the left.
Spawn x = -60	ltr-net-center*.mp4 (16:35–16:43 PT).	Not the look. Confirmed drop is further left: x = -200.

Reference tapes (mtimes PT = UTC+2): ltr-net-left3.mp4 15:59 PT (first tape they pointed at); ltr-firstcave-neg200.mp4 17:41 PT (restore; confirmed 17:43 PT).

5. Right → left (RTL) — RECIPE, not live-confirmed

Not taped tonight. Do not claim a live look.
Same cave, same 8B CA (keep the Z-fan). Change only the travel float: BA 00 00 80 3F (+1.0 LTR) → BA 00 00 80 BF (-1.0 RTL).

Spawn should be on the **right** if you want the band to start on the player's right and sweep left. Try **x=+200** (past the right edge, same saturation logic as -200 on the left) or **x=+55** (right sideline). Which of those two looks better is **UNKNOWN** — not taped tonight. LTR +1.0 is the confirmed sign, so -1.0 is the honest RTL recipe. If the wall still goes LTR after the float swap, the old exe is still loaded or the write missed. Restart the client and re-hash. Do not invent the post-edit sha256; hash after you write.

6. Up → down and down → up — VANILLA AXIS

Vanilla (no LTR cave) already travels baseline↔net. Screen UTD = top → bottom (toward camera). Screen DTU = bottom → top (away from camera). Travel is **client Z / script Y**.

Sign of vanilla dir.z comes from the attacker, not from planted xyz

dir.z = attacker-related Z from 0x524C30. ShotRot=0 $\Rightarrow$ dest does not steer. Planted Y moves the drop along the court; it does **not** flip heading. Dummy-4 world position was not dumped.

Coordinate notes (origin-findings + guardian-shot-ai-re): $-Y$ = player / near baseline = $-client$ Z. So **$-client$ Z is toward the player / down the screen**. A forced cave that ignores attacker would set:

```
UTD (toward player / toward camera), IF the coordinate mapping holds:
    dir.z = -1.0          ; B9 00 00 80 BF    mov ecx, 0xBF800000
    dir.x = old X-spread ; edx unchanged    wall stays full-width
    dir.y unchanged
DTU (away from camera): same, dir.z = +1.0 ; B9 00 00 80 3F
```

That sign is **inferred from the coordinate system**, not taped tonight. If the live wall goes the wrong way, flip dir.z. Exact 30-byte UTD/DTU cave encoding (jne displacement, nops) was **not assembled tonight**. Do not invent a hash for a UTD/DTU exe. Register-level recipe only: same site jmp to the same padding; skillId 27 only; leave edx alone; mov ecx to ± 1.0 .

If you cannot prove the sign live, write this and stop

Vanilla already travels this axis. Which way depends on attacker Z. A forced cave would set dir.z to ± 1.0 and keep the X-spread. Sign must be proven live. Whether flipping attacker is enough for vanilla UTD vs DTU without a cave is UNKNOWN.

To test vanilla UTD/DTU **without** a cave: revert to the LAA-only backup (sha eebd71a1...), keep skillId 27, plant Y on the far baseline (script y = +115 or +150) or the near baseline (y = -115 or -150), and watch which way the wall walks. That shows what dummy-4's attacker Z is doing. It does not prove you can choose the sign.

Four-direction recipe matrix

Same 14-byte site (jmp 0x6E5C58 + nops) and same return (jmp 0x4EA125) for any cave. Vanilla UTD/DTU needs no cave. Do not invent a hash for an exe you have not written.

	LTR CONFIRMED	RTL RECIPE	UTD vanilla-axis	DTU vanilla-axis
dir.x after cave	+1.0 (BA ... 3F)	−1.0 (BA ... BF)	old X-spread (leave edx)	old X-spread (leave edx)
dir.z after cave	old X-spread (8B CA)	old X-spread (8B CA)	−1.0 forced, or attacker Z	+1.0 forced, or attacker Z
Foam	Vertical band + Z-fan	Vertical band + Z-fan	Full-width wall along X	Full-width wall along X
Travel	+X left → right	−X right → left	−Z toward player (if coord notes hold)	+Z toward far baseline (if coord notes hold)
Spawn to see it	x=−200 (confirmed)	try x=+200 or +55	y=+150/+115 shows drop; does not steer	y=−150/−115 shows drop; does not steer
Exe tonight	live cf551df8...	not written / not hashed	LAA-only eebd71a1... or unbuilt cave	LAA-only eebd71a1... or unbuilt cave
Live tape	YES 17:43 PT	NO	vanilla axis only; sign UNKNOWN	vanilla axis only; sign UNKNOWN

7. SHA-256 (verified on this box 18 Aug 2026 ~19:40 PT)

All four files are 3 801 088 bytes.

File	Role	SHA-256
client/FantaTennis.exe	LIVE first LTR cave (8B CA, +1.0). Use this for LTR.	cf551df8bd42f32676f8b01e54496a1b74473c90ddfa1354288f6545dea92f7c
client/FantaTennis.exe.seawave-ltr.bak	LAA-only backup (pre-cave). Site is the original 14-byte stores. Cave region is zeros. VirtualSize 0x2E4C58. LAA 0x123.	eebd71a1b19eca60101a195c49999d29600e6cdd6302c4a6001fb98043f91162
evidence/.../FantaTennis.exe.ltr	DO NOT USE. Z-spread-zero cave (31 C9).	967cc05871330521c9c7dc0ade4f9f5989f9ce5797d81362174c10c985ff71f3f
client/FantaTennis.exe.official	Stock official (no LAA, no cave). Characteristics 0x103. Reference only.	5477f0827acae66976403aecd2e9ebffeb4fa28da1fedae5f9541ec25e336c31

Official vs LAA-only differ by PE IMAGE_FILE_LARGE_ADDRESS_AWARE (0x20). Launch helper scripts/apply-ft-client-patches.py is an official-test stub: it **only reapplies LAA** and skips invite caves. It does not write the cave region. It must not wipe the LTR cave — the current stub does not. An exe change still needs a **client restart**. Servers do not need a restart for the cave.

apply_ltr_patch.py / bytes.txt are stale

They still encode the Z-spread-zero cave (31 C9). Re-running that script on a clean exe installs the wrong look. first-ltr-restore.txt is the record of putting 8B CA back. Live exe already has the first cave. Do not “fix” it with the apply script.

8. Scripts, HP pad, waiting-room arm

Live fire path is game-server/target/dist/scripts/. Source copies live under game-server/src/main/resources/scripts/.

Path	Role
dist/scripts/command/wavetest.js	−wavetest. Waiting room → writes /tmp/jftse-wavetest.arm, chat “Armed. START a guardian match; volley fires after the intro.” In-match → fires at once. Guardian only.
dist/scripts/event/3_wavetest_autopad.js	On MP_GAME_ANIM_SKIP_END: pad HP, then if the arm file exists, delete it and fire the same volley.

At write time (~19:40 PT) the live scripts are not the confirmed −200 take

src command + src autopad: five steps all x: −500 “net line, further left (−500)”. dist command + dist autopad: mixed experimental steps (x: −500, y: ±500, y: −200, y: 0). To reproduce tonight’s confirmed LTR look: put **both** dist files back to five copies of { attacker: 4, x: −200, z: 0, y: 0 }, then −reloadScripts. Events stay stale in memory otherwise. These two scripts do not replace guardian-phase/10.

HP pad 30000. Client HP is a **short**; 99999 overflows. Iterate playerBattleStates with .iterator() (not .get(i)), setMaxHealth(30000), currentHealth.set(30000), setDead(false), then S2CMatchplayDealDamage(0, 30000, 0, 1, 0.0, 0.0) (heal skill 1). The bar may still draw 200/200; popups are real hits. Solo Testmon otherwise dies in about 15 s. Morning 9-drop popups (WAVE 1 = 9708, WAVE 4 = 9042) were vanilla-axis hits, not re-measured for LTR.

9. Ops (tonight's session)

Item	Value
Client	FantaTennis.exe via Proton GE-Proton11-1 (proton-ge11/version).
Auth / game / relay / chat	TCP 5894 / 5895 / 5896 / 5897.
AC	Real AC on TCP 3724 (docker-compose 3724:3724). Need a live TCP session, not a leftover Wine process.
Splash "Initializing..."	Leftover Wine or no AC TCP. Healthy: AC channelActive within ~15 s, register result:0, heartbeats ~10 s.
Account	test / Testmon (Lv 01) used tonight.
Guardian room	Lock the 3 empty slots before START enables.
Skip intro	~top-right. DISPLAY :10 1280×800 working click tonight: (959, 62) — session-specific.
START vs Linux dock	Dock covers the bottom UI. Click the top edge of START. Tonight ~(821, 655) on 1280×800. A low click opens the host browser.
Exe vs scripts	Cave change = restart the client. Script change = -reloadScripts. Launch helper may only set LAA.

10. UNKNOWN / not re-verified

- RTL live look. Not taped tonight.
- Forced UTD/DTU cave signs. Coordinate notes imply dir.z = -1.0 toward the player and +1.0 toward the far baseline. Must be proven live.
- Whether flipping attacker is enough for vanilla UTD vs DTU without a cave.
- Fielditem_INI3.set was not decrypted. Data-clone is still a dead end for travel.
- LTR knockback feel. 0x4F1C91 X-zero is still in the exe.
- Whether x=-500 is "more left but same look" beyond saturation. Not confirmed as the take.
- ShotSpeed / per-crest +index*35 and dest-Y force (VA 0x4EA14F) come from the morning RE note; not re-read in a debugger tonight.
- Which Proton prefix Proton will load from a different helper. Confirm live exe sha256 after launch.

Related notes

Morning origin-findings	JFTSE-wavetest/docs/seawave-origin-findings-2026-08-18.md. xyz plants the drop; vanilla travel is baseline↔net. "LTR is not possible" is superseded.
First LTR cave addendum	docs/seawave-ltr-first-cave-2026-08-18.md — confirmed 8B CA cave + (-200, 0, 0).
Client RE	evidence/seawave-ltr/README.md — mover, ShotRot, skillId 27 first-frame. Written before live LTR confirmation; travel-unverified paragraph is stale.
bytes.txt / apply_ltr_patch.py	STALE. Still encode 31 C9. first-ltr-restore.txt is the restore record.
This box's scripts	Live fire = dist copies. At write time they are not the -200 take — restore before a confirmation replay.

How to fire each direction

One page. Pick a direction, set the matching lever, fire skillId 27, watch travel. If a box fails, stop — do not tune spawn to hide a wrong cave.

Common (every direction)

■	Room / UI Guardian Battle Mode. Lock 3 empty slots. START ~top edge (~821, 655 on 1280×800 :10). Skip intro ~top-right. Dock covers the bottom.
■	Scripts ~reloadScripts after every edit. Wait for "Scripts reloaded". Waiting room ~wavetest = arm. In-match ~wavetest = fire now. Guardian only.
■	HP / skill HP pad 30000 (client HP is a short). Packet skillId 27, attacker 4. No new skill id. A FieldItem clone will not pick up the cave and will not go sideways.
■	Exe / record Cave change = restart client. Launch helper must not wipe the cave (current stub only sets LAA). Hash the live exe after boot. Record before START.

A. LTR — CONFIRMED

■	Exe sha256 = cf551df8bd42f32676f8b01e54496a1b74473c90ddfa1354288f6545dea92f7c. File 0x2E5C58 has 8B CA at cave+6, then BA 00 00 80 3F. Not 31 C9. Site 0x0EA115 = E9 3E BB 1F 00 + nops.
■	Spawn Five steps { attacker: 4, x: -200, z: 0, y: 0 } "net center, further left". Not -500, not +200, not -60.
■	Watch Vertical band, walks left → right, foam may show at both baselines. Labels xyz=(-200, 0, 0). Thin net-line = Z-zero cave. Sideline-spanning wall walking baseline↔net = vanilla (no cave).

B. RTL — RECIPE, not taped tonight

■	Cave Same as LTR except BA 00 00 80 BF (-1.0). Keep 8B CA. Restart client. Hash will not be cf551df8... — hash after you write; do not invent it.
■	Spawn / watch Start on the right: try x=+200 or x=+55 (which is better is UNKNOWN). Vertical band, walks right → left. Mark the tape RECIPE.

C. UTD — VANILLA AXIS (toward player / toward camera)

■	Exe No LTR cave: LAA-only backup sha eebd71a1b19eca60101a195c49999d29600e6cdd6302c4a6001fb98043f91162. Or a forced cave dir.z = -1.0, keep X-spread (encoding not assembled tonight). Sign inferred, not taped.
■	Spawn / watch Planted Y moves the drop, not the heading. Far-baseline drop (step.y = +150 or +115) to see it walk toward you. Full-width wall, screen top → bottom. If it walks the other way, attacker Z is the other sign — that is the UNKNOWN. Do not claim a forced-cave sign without a tape.

D. DTU — VANILLA AXIS (away from camera)

■	Exe / spawn / watch Same as UTD, opposite dir.z (+1.0 if you force it). Plant on the near baseline (step.y = -150 or -115) to see it walk away. Full-width wall, screen bottom → top. Whether flipping attacker (dummy 4 vs 5, or a far-side guardian) is enough without a cave is UNKNOWN.
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If the foam matches the first LTR tape, stop. Do not then change X, zero Z, or swap +1.0 for -1.0 "just to see." That is how tonight's LTR look was lost for an hour.

Written 18 August 2026, Europe/Berlin (PT), after Thewind's 17:43 PT LTR confirmation. Sources: live client/FantaTennis.exe + backups, S2CMatchplayUseSkill.java, dist/src wavetest.js, evidence/seawave-ltr/{README.md, bytes.txt, first-ltr-restore.txt, apply_ltr_patch.py}, docs/seawave-ltr-first-cave-2026-08-18.md, JFTSE-wavetest/docs/seawave-origin-findings-2026-08-18.md, FieldItem_Skills_Ini3.xml SeaWave ID 27, notes/guardian-shot-ai-re.md, scripts/apply-ft-client-patches.py, proton-ge11/version.